

KERALA AGRICULTURAL UNIVERSITY

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SPICES AND CONDIMENTS

CARDAMOM (*Elettaria cardamomum*)

The habitat of small cardamom is the evergreen forests of Western Ghats. It is grown in areas where the annual rainfall ranges from 1500-4000 mm with a temperature range of 10-35 °C and an altitude of 600-1200 m above MSL.

Cardamom is generally grown in forest loam soils rich in available phosphorus and potassium. The crop is raised mainly on well drained, deep, good textured soils rich in humus.

Varieties

ICRI-1, ICRI-2, PV-1 and PV-2. ISSR Vijetha is resistant to Katte disease and is recommended to moderate rainfall with moderate to high shaded mosaic (cardamom mosaic virus) infected areas. IISR Avinash is resistant to rhizome rot and is highly suitable for planting in valleys.

Cultivars

Malabar: Suitable for areas from 600 to 1200 m elevation

Mysore: Suitable for areas from 900 to 1200 m elevation

Vazhukka: Suitable for areas from 900 to 1200 m elevation

Propagation

Cardamom can be propagated vegetatively and by seedlings.

Vegetative propagation

In Kerala, vegetative propagation is commonly practiced. For vegetative propagation, rhizomes with an old shoot and a sprout are used. Plants propagated vegetatively come to bearing one year earlier than the seedling propagated plants and are true to type.

Rhizome multiplication

This may be taken up from the first week of March to the first fortnight of October. The site is selected in open, gently slopping and well-drained areas near a source of water. Trenches of 45 cm width, 45 cm depth and convenient length are taken across the slope or along the contour 1.8 m apart. They are filled with equal quantity of humus rich topsoil, sand and cattle manure. Uproot a part of the high yielding disease free mother clump identified in the plantation. Trim the roots and separate the suckers so that the minimum planting unit consists of one grown up tiller and a growing young shoot. Plant them at a spacing of 1.80 m x 0.60 m in filled up trenches. Provide sufficient mulch and stake each planting unit. Provide overhead pandal as in the case of seedling nursery and remove shading material with onset of monsoon rains. Provide irrigation once in a fortnight and adopt necessary plant protection measures. Apply fertilizers @ 100:50:200 kg ha⁻¹ N:P₂O₅:K₂O in six splits at an interval of two months. Apply neem cake @ 100-150 g/plant along with fertilizers. On an average, 20 to 30 suckers / initial planting unit can be produced within one year of planting. Care should be taken to identify and collect mother clumps only from areas totally free from 'katte' disease.

Seed propagation

Vegetative propagation has the disadvantage of spreading the 'katte' disease, which is of viral origin. This disease is not transmitted through seeds. Hence in areas where the disease is widespread, it would be safer to use seedlings for propagation.

Nursery

Ripe capsules of the desired cultivar are collected from high yielding plants during September-October. The seeds are extracted by gently pressing the capsules. In order to increase the germination percentage, seeds can be treated with concentrated sulphuric acid or nitric acid for not more than two minutes. The extracted seeds are washed in cold water four times to remove the mucilaginous coating. The washed seeds are drained and mixed with ash and allowed to dry in shade for 2 or 3 days. The seeds should be

sown in the nursery within a fortnight. Sowing in September is the best for high germination. Sowing during southwest monsoon and winter should be avoided.

When it becomes necessary to store the seeds, it is advisable to store them in capsule form. It can be preserved in this form for one month, without deterioration of viability. Polythene lined gunny bags can be used for this.

In Kerala and Tamil Nadu, 18 month old seedlings are used for planting. The seeds are sown in primary nursery from where the young seedlings are transplanted to a secondary nursery and maintained for one year before planting in the main field.

Primary nursery

The nursery site is selected in open, well-drained areas, near a water source. The land is dug to a depth of 30 cm, cleared of all stubbles and stones; and clods are broken. Beds of size 6 m x 1m x 0.3 m are then prepared. Jungle soil is spread in a thin layer over the nursery bed. Seeds are sown on the bed in lines. For an area of 1 m², 10 g of seed is required. Sixty grams of seeds will be required for a nursery bed of 6 m². The seeds are covered with a very thin layer of fine soil. The nursery bed is mulched with dry grass. Potha grass (*Grenetia stricta*) commonly seen in high range areas is a suitable material for this purpose. Grass is spread to a thickness of about 2 cm. Paddy straw can also be used for mulching. After sowing, beds have to be watered every day in the morning and evening. The mulch should be removed on commencement of germination. The seedlings have to be protected by providing shade pandals. Regular watering, weeding and protection from pests and diseases are to be attended to. During June-July, seedlings from the primary nursery are transplanted to the secondary nursery.

Secondary nursery

After preparing the site properly, form nursery beds of 6 m x 1m x 0.3 m. Mixing of well decomposed cattle manure and wood ash with the top layer of the soil will help the seedlings to establish well and to grow vigorously. During June-July, the seedlings from the primary nursery are transplanted at a spacing of 25-30 cm. Shade pandals should be provided before transplanting. Overhead pandals or individual pandals for each bed may be erected. Mulching the bed with dry leaves will help to conserve soil moisture. Regular watering during dry months, weeding, application of fertilizers, control of pests and diseases and mulching are the essential operations for the maintenance of the secondary nursery. One month before uprooting, the pandal should be removed to encourage better tillering.

Polybag nursery

Polybags can be used for raising secondary seedlings. For such nurseries, seeds are to be sown in beds in primary nurseries in September and transplanted to polybags in December-January. These seedlings would be ready for planting in June-July. In this case, nursery period could be reduced by 6 to 7 months.

Soil treatment in nursery

It is recommended that the primary and secondary nursery soil may be drenched with formalin 2 per cent solution and covered with polythene sheets for three days. Planting should be taken up only 15 days after treatment to avoid phytotoxicity.

Control of pests and diseases in the nursery

Rhizome weevil (*Prodiectes haemeticus*)

This is a serious pest in the secondary nursery especially where seedlings are raised continuously year after year. The grubs feed on the rhizome and basal portion of the stem. This results in drying of leaves and breaking of stem at the base. Drenching the nursery beds with chlorpyrifos at 0.04 per cent can control the pest.

Shoot fly (*Formosina flavipes*)

The pest is observed in the nursery during January to May. Dead heart or decay of the central spindle is the external symptom. Spraying quinalphos 0.05 per cent is recommended for the control of the pest.

Shoot borer (*Conogethes punctiferalis*)

The caterpillar bores into the stem and feeds on the internal contents. This results in the decay of the central spindle and production of dead heart. Faecal matter of the caterpillar can be seen coming out through the holes. Spraying with quinalphos 0.05 per cent, carbaryl 0.1per cent or dimethoate 0.05 per cent is recommended against the shoot borer.

Nematodes

Nematodes are observed as serious pests in cardamom nurseries. Roots of cardamom seedlings are infested mainly by root knot nematode (*Meloidogyne incognita*). Lesion nematodes (*Pratylenchus*) are also

seen in cardamom roots and soils. The main symptoms of nematode infestation are galls on the root tips, profuse tillering, stunted and weak tillers, yellowing and drying of leaves and production of narrow, brittle and abnormal leaves. Treatment of soil as detailed above is an effective method to control nematode. Pruning of infested root tips before planting is also recommended.

Damping off

This disease is caused by *Pythium vexans* and *Rhizoctonia solani*. Infection is observed at the collar region. Provide good drainage, and spray and drench the nursery with 1 per cent Bordeaux mixture or 0.2 per cent copper oxychloride.

Nursery leaf spot

This disease is caused by *Phyllosticta eletariae*. Pale specks appear on the leaf lamina, which dry up and become papery white. Spraying the plants with mancozeb 0.25 per cent at fortnightly intervals is effective in controlling the disease.

The other diseases are *Sphaceloma* leaf spot, *Cercospora* leaf spot, rust and sooty mould.

Main field planting

Cardamom plantation is raised in forests under the shade of tall trees. For raising a new cardamom plantation, the undergrowth of bushes is cleared. When open areas like marshy valleys and grasslands are selected for raising new plantation, shade trees have to be raised before planting cardamom seedlings. The quick growing shade trees like dadap (*Erythrina lithosperma*) is generally used for this purpose. Cuttings of this tree are used for planting. But this tree is a host of root knot nematode, which infests cardamom. Other quick growing trees like *Albizia* can also be used. Useful trees like jack and eucalyptus can be used along with red cedar, wild nutmeg, *kurangatti* etc.

Spacing

Mysore and Vazhukka: 2 m x 2 m to 3 m x 2 m depending on the fertility of the soil Malabar: 1.5 m x 1.5 m to 2 m x 2 m depending on the fertility of the soil.

The recommended size of pits is 60 cm x 60 cm x 35 cm. The pits are filled with rich topsoil at least two months in advance of planting the seedlings. Application of well decomposed FYM or compost or leaf mould and 100 g of rock phosphate with the topsoil in the pit will help in proper establishment and quick growth of plants. If the selected site is a hill slope, terraces may be formed before digging pits.

Planting can be done with the commencement of southwest monsoon, before the heavy rains. A small pit may be formed inside the pit by scooping out soil at the centre of the pit for planting seedlings. The soil may be put just to cover the rhizomes. Care should be taken to ensure that the rhizomes do not go deep into the soil.

Cultural operations

A regular schedule of cultural practices consisting of weeding, mulching, trashing, shade regulation, fertilizer application, irrigation, etc. will have to be undertaken.

Sufficient mulch should be applied at the base of the plant during December to reduce the ill effects of drought during summer months and to conserve soil moisture. Sickle weeding is essential which has to be carried out frequently depending upon the intensity of weeds. Forking is necessary in hard soils, which is to be carried out in October- November.

Trashing (removal of old and dried shoots, leaves and dried panicles) should be taken up once in a year during June-July, with the commencement of monsoon. This will help to prevent the spread of diseases and expose the panicles to easy visit by honeybees.

Soil conservation measures, maintenance of drainage channels and such other operations may be taken up promptly.

Manuring

Application of organic manures such as FYM, cowdung or compost @ 5 kg / plant or neem cake @ 1-2 kg / plant may be done during June-July. The present recommendation of nutrients for cardamom in Kerala is N:P₂O₅:K₂O @ 75:75:150 kg ha⁻¹. The fertilizers may be applied in two split doses, before and after the southwest monsoon, in a circular band of 20 cm wide and 30-40 cm away from the base of the clumps, and mixed with soil.

Shade

Since inadequate as well as excessive levels of shade are harmful to the crop, regulation of shade is

inevitable. There should be sufficient shade to protect cardamom plant during the hot season. By regulating the shade before the monsoon, more light becomes available to the plant during the rainy season. Red cedar or chandana-vempu (*Toona ciliata*) is an ideal shade tree. It sheds the leaves during rainy season and thus provides natural shade regulation. Some of the other shade trees are kurangatti (*Acrocarpus fraxinifolius*), vellakil (*Dysoxylum malabaricum*) and thelli (*Canarium strictum*).

Bee-keeping for better pollination

The main pollination agent in cardamom is honeybee (*Apis cerana indica*). Maintaining four bee colonies per hectare during the flowering season is recommended for increasing fruit set and production of capsules.

Harvesting and processing

Cardamom plants normally start bearing capsules from the third year of planting. Picking is carried out at an interval of 30 days. After harvest, cardamom capsules are processed. Cardamom capsules with green colour fetch a premium price in foreign countries. Hence emphasis has to be given on the preservation of green colour during curing and subsequent storage. Capsule should be processed within 24-36 hours after harvest to prevent deterioration. By curing, the moisture of green cardamom is reduced to 8-12 per cent at an optimum temperature so as to retain its green colour to the maximum extent. Harvesting is done almost round the year in Kerala with the peak period from August - December.

Artificial drying

Processing of capsules is done in specially built curing houses. The harvested capsules are washed in water to remove dust and soil particles. Then they are spread on wire net trays in curing chamber. Burning firewood in the iron kiln produces heat required for drying. The heat thus produced is passed through pipes made of galvanized iron sheets. The process of drying takes about 18-24 hours, depending on the ambient temperature. The capsules are spread thinly in the wire net trays and stirred frequently to ensure uniform drying. They are initially heated at 50 °C for the first 4 - hours and heat is then reduced to 45 °C by opening ventilators and operating exhaust fans till the capsules are properly dried. Finally the temperature is raised to 60 °C for an hour.

The dried capsules are rubbed on wire mesh to remove the stalk and dried portion of flower from the capsules and then graded according to size by passing through sieves of sizes of 7, 6.5, 6 mm etc. The graded produce is stored in polythene lined gunny bags to retain the green colour during storage and also to avoid exposure to moisture.

A relatively new innovation in the curing procedure is blanching by soaking the fruits in 2.0 per cent washing soda for 10 minutes prior to drying. This inhibits colour loss during drying operation and extends colour retention during subsequent storage from three months to ten months.

Sun drying

Capsules are dried directly under sunlight for five to six days or more. Frequent turning is done. This method can result in surface blemishes and may not give an attractive green colour. This method is practiced if the cultivar yields fruits that turn yellow before they are ready for picking and where facilities for green curing are not available.

Bleaching

A proportion of the crop is bleached after sun drying by exposing the capsules to fumes from burning sulphur to get uniform colour and appearance. Steeping capsules in a dilute solution of potassium metabisulphite induces a slight improvement in keeping quality.

Oleoresin

Solvent extraction of ground spice yields 10 per cent oleoresin. Cardamom oleoresin dispersed in salt, flour etc is used for flavouring food. One kilogram of oleoresin replaces 20 kg ground spice.

Decorticated seeds / seed powder

Decorticated seeds command a lower price due to rapid loss of volatile oil during storage and transportation. Seed powder is marketed to a limited extent.

Control of pests and diseases in the plantation

Cardamom thrips (*Sciothrips cardamomi*)

This insect is a serious pest of cardamom. It colonizes and breeds in unopened leaves, leaf sheath, flower bracts and flower tubes. It lacerates and feeds on the exuding sap from the aerial parts. Infestation on the panicle and flower buds results in stunted growth of panicles, shedding of flower buds and warty growth on the surviving capsules. The infested capsules are light in weight, inferior in quality and fetch very low price in the market. Since the pest population is high during dry months from December to May, pesticide

application during this period is important. Four sprayings of insecticide during this period is recommended. Insecticide application can be skipped during rainy months of June and July. Three more sprayings are to be given during the period from August-November. Any of the following insecticides are recommended for thrips control.

Quinalphos 0.05 per cent, phosalone 0.07 per cent, dimethoate 0.05 per cent.

Shoot/capsule borer (*Conogethes punctiferalis*)

It is a serious problem to cardamom growers of Kerala, Tamil Nadu and Karnataka. At the early stage of the crop, the caterpillars of this yellow coloured moth bore into the core of the aerial stem resulting in the death of central spindle, which appears as characteristic dead hearts.

At the time of flowering, when the caterpillars attack the panicles and spikes it may lead to flower shedding and drying up of the attacked portions. At a later stage of the crop, the caterpillars bore into the capsules, feed on the seeds and make them hollow. The presence of excreta at the region of attack indicates presence of the caterpillars in the pseudostem, inflorescence and pods.

Pest infestation is pronounced in three seasons viz. January-February, June and September-October.

Management

Later stages of larvae bore into the pseudostem and remain there. Insecticides sprayed at this time may not give adequate control of the pest. For an effective management of the pest, the insecticides have to be targeted on early stages of the larvae, which are usually present within 15-20 days after adult emergence in the field.

Leaf eating caterpillars

There are 10 species of caterpillars feeding on cardamom leaves. Out of these, seven species are hairy and appear in large numbers during certain seasons causing extensive defoliation. For controlling the leaf caterpillars, mechanical collection and destruction and spraying of any contact insecticide are recommended.

Cardamom whitefly [*Kanakarajiella (Dialeurodes) cardamomi*]

It is a serious pest in cardamom growing tracts of Kerala. The adult is a small soft bodied insect, about 2 mm long and having two pairs of white wings. The nymphs are elliptical and pale green. The nymphs secrete sticky honeydew, which drops on to lower leaves. On these, black sooty mould develops, which interrupts photosynthesis of the leaves.

Management

The flies are attracted towards yellow colour. So metal sheets painted yellow and coated with sticky materials, such as castor oil or poly-venyl butanol would serve as traps. By placing such yellow sticky traps between rows of cardamom plants, population of adults can be monitored and adults trapped to some extent. Nymphs are effectively controlled by spraying the lower surface of leaves with a mixture of neem oil (500 ml) and triton (500 ml) in 100 litres of water. Acephate 0.1 per cent is effective. The spray may be repeated two or three times at 15 days interval.

Cardamom root grubs (*Basilepta fulvicorne*)

The grubs of a small, greenish blue beetle cause damage. The grubs are short, stout, pale white in colour and often assume a shape resembling 'C', which feeds on cardamom roots. The symptoms start as yellowing of leaves, which later result in the drying up and death of the plant.

Management

Collect the beetle with hand nets or sticky traps at the time of mass emergence (March-April and August-September) and destroy. Early stages of the grub which are usually present in soil during May-June and September-October can be controlled by drenching chlorpyrifos 0.04 per cent @ 3-4 litre per clump 10-15 cm around the plant.

Cardamom scale (*Aulacaspis* sp.)

This scale insect is found on the lower surface of leaves, leaf sheath, panicles and fruit stalk. As a result of damage, capsules get shrivelled, panicles become dry and the leaves become yellow. The pest is mostly seen during summer months.

Nematodes (*Meloidogyne* sp.)

Root knot nematodes are the most common nematode species associated with cardamom plantations. Common symptoms are necrosis of leaf tips and margins, narrowing of leaves, thickening of veins,

reduction of internodal length and rosetting. Roots branch heavily and galls appear on them. Plant becomes highly stunted.

Management

Frequent change of nursery beds will help to reduce nematode infestation in nurseries.

Diseases

Katte or mosaic

This is a virus disease, which is transmitted by the banana aphid, *Pentalonia nigronervosa*. The symptoms consist of discontinuous stripes of light green colour running almost parallel to each other from the mid-rib to the margin of the leaves, which form a mosaic pattern. On young shoots, such stripes are seen on the leaf sheath also. The infected clumps will be smaller in size with fewer tillers.

Management

Eradication of the source of inoculum by destroying infected plants and destruction of the vector by insecticide application are effective. Regular application of insecticide against cardamom thrips controls the aphids also. Avoid using katte-infected rhizome for planting.

Destruction of plants showing symptoms of the disease should be done promptly once in two months. Removal of all alternate hosts of virus is also recommended.

Azhukal

This is a fungal disease caused by *Phytophthora* sp. occurring during the rainy season. It affects the leaves, tender shoots, panicles and capsules. On the infected leaves, water soaked lesions appear first and rotting and shedding of leaves along the veins occur thereafter. The infected capsules become dull greenish brown and decay. This emits a foul smell and subsequently shed. Infection spreads to the panicles also.

Management

Trashing and destruction of the infected parts should be done as a phytosanitary measure just prior to the onset of southwest monsoon. Remove the trash (dried leaves and leaf sheaths) from the basal region of the plant to the extent possible.

Spray the shoots with 1 per cent Bordeaux mixture with adhesive (rosin soda or any other sticker) by the commencement of the monsoon and continue the spraying operation two or three times up to November-December according to the intensity of the disease and rainfall. Give a copious spray to the panicle with 1 per cent Bordeaux mixture @ 3 l per plant during July-August when the disease intensity is maximum.

Trichoderma can be used along with cowdung for controlling this disease.

Clump rot or rhizome rot

This disease is caused by *Pythium aphanidermatum*, *P. vexans*, *Rhizoctonia solani* and *Fusarium oxysporum*. The affected shoots become brittle and easily break off from the rhizome at the bulbous base.

Management

Drench with 0.2 per cent copper oxychloride (2-3 litre/ plant) and repeat this two times at monthly intervals.

As a bio-control measure, inoculate seedlings with native arbuscular mycorrhiza, *Trichoderma* and *Pseudomonas fluorescens* at the time of planting in the nursery and main field, and apply during pre-monsoon period in established plantations (see the chapter on biocontrol agents against plant pathogens).

Leaf blotch disease

The fungus *Phaeodactylium venkatesanum* causes this disease. The disease is characterized by the appearance of large blotches of irregular lesions with alternating shades of light and dark brown necrotic tissues. This is mainly observed on mature leaves. On the lower surface of the lesions ash coloured white superficial growth of the fungus appears during moist weather conditions.

Management

The fungicides, Bordeaux mixture (1 per cent), mancozeb (0.3 per cent) and carbendazim (0.05 per cent) are effective in controlling the disease.

Chenthall disease

Chenthall disease is characterized by the appearance of rectangular linear reddish brown lesions mainly on the lower surface of the leaves. The lesions are clearly visible even on dried leaves. The incidence of the disease appears to be more severe in areas, which do not have proper shade. Even though

Corynebacterium and *Colletotrichum gloeosporioides* have been isolated from the infected leaves, the pathogenicity of these organisms could not be established.

Management

Providing adequate shade is the only measure recommended pending confirmation of etiology of the disease.

Waiting period of insecticide / fungicide

Quinalphos	30 days
Mancozeb	30 days

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CINNAMON (*Cinnamomum zeylanicum*)

Cinnamon grows in areas up to an altitude of about 1800 m. Humid tropical evergreen rain forest conditions favour the best growth of cinnamon. Well-drained, deep sandy soil, rich in humus is suitable for the crop. Avoid marshy areas and hard laterites.

Varieties

Navasree, Nithyasree and Sugandini.

Seeds and sowing

Cinnamon is usually propagated through seeds. Sow seeds immediately after harvest on raised beds. Pot seedlings when they are six months old.

Vegetative propagation

For raising cinnamon from cuttings; semi hardwood cuttings of about 10 cm length with 2 leaves are taken and dipped in IBA 2000 ppm and planted either in polythene bags filled with sand or a mixture of sand and coir dust in the ratio 1:1 or in sand beds raised in a shaded place. The cuttings in polythene bags must also be kept in a shaded place or in a nursery. The cuttings are to be watered regularly 2-3 times a day for maintaining adequate moisture and prevent wilting. Rooting takes place in 45-60 days. The well rooted cuttings can be transplanted to polythene bags filled with potting mixture and maintained in a shaded place and watered regularly.

Air Layering

Air layering is recommended for all cinnamon nurseries. It is done on semi hardwood shoots. A ring of bark is removed from the semi hardwood portion of the shoot and a rooting hormone (IBA 2000 ppm or IAA 2000 ppm) is applied on the portion where the bark has been removed. Moist coir dust or coir husk is placed around the region where the hormone has been applied and is secured in position by wrapping with a polythene sheet of 20 cm length. This would also avoid moisture loss. Rooting takes place in 40-60 days. The well rooted air layers are separated from the mother plant and bagged in polythene bags filled with potting mixture and kept in a shaded place or nursery by watering the plants twice daily.

Planting

Select seedlings with green leaf petioles. Plant seedlings in the main field when they are 1-2 years old with the commencement of southwest monsoon. Planting is done in pits of size 60 cm x 60 cm at a spacing of 2 m x 2 m. Dig the pits sufficiently early to allow weathering. Fill the pit with leaf mould and topsoil before planting.

Manuring

Apply $N:P_2O_5:K_2O$ @ 20:20:25 g per seedling in the first year and double this dose in the second year. Cattle manure or compost at 20 kg per plant per annum may also be applied. Increase the dose of $N:P_2O_5:K_2O$ gradually to 200:180:200 g per tree per year for grown up plants of 10 years and above. Apply organic manures in May-June and fertilizers in two equal split doses, in May-June and September-October.

Aftercultivation

Weed regularly in the early stages of growth. Irrigate the seedlings till they get established, if there is long drought period. Prune plants when they are 2-3 years old at a height of 15 cm above ground level. Cut the side shoots growing from the base to encourage growth of more side shoots till the whole plant assumes the shape of a low bush.

Harvesting and curing

The plants will be ready for harvest in about 3 years after planting. Harvesting is done during two seasons, the first in May and second in November. The correct time for cutting the shoots for peeling is determined by noting the sap circulation between the wood and corky layer. Peelers can judge this by making a test cut on the stem with a sharp knife. If the bark separates readily, the cutting is taken immediately. Stems measuring 2.0 to 2.5 cm in diameter and 1.5 to 2.0 m length are cut early in the morning and twigs and leaves are detached. The outer brown skin is first scrapped off and the stem is rubbed briskly to loosen the bark. Two cuts are made round the stem about 30 cm apart and two longitudinal slits are made on opposite sides of the stem. The bark is separated from the wood with curved knife. The detached pieces of bark are made into compound quills. The best and longest quills are used on the outside while inside is filled with smaller pieces. The compound quills are rolled by hand to press the outside edges together and are neatly trimmed. They are dried in shade as direct exposure to sun can result in warping. The dried quills consist of mixture of coarse and fine types and are yellowish brown in colour.

The quills are graded as Fine or Continental, Mexican and Hamburg or Ordinary. The Fine consists of quills of uniform thickness, colour and quality and the joints of the quills are neat. Mexican grades are intermediate in quality. The Hamburg grade consists of thicker and darker quills. The lower grades are exported as: (a) *Quillings*: The broken lengths and fragments of quills of all grades are bulked and sold as quillings; (b) *Featherings*: This grade consists of the inner bark of twigs and twisted shoots that do not give straight quills of normal length.

Chips: This includes the trimmings of the cut shoots, shavings of outer and inner bark, which cannot be separated, or which are obtained from small twigs and odd pieces of thick outer bark.

Oleoresin

Cinnamon oleoresin is prepared by extracting cinnamon bark with organic solvent. Oleoresin yield varies from 10 to 12 per cent. The oleoresin is dispersed on sugar, salt and used for flavouring processed foods.

Cinnamon bark oil

A pale yellow liquid possessing the delicate aroma of the spice is obtained by steam distillation of quills (0.2 to 0.5 per cent). Its major component is cinnamaldehyde (55 per cent) but other components like eugenol, eugenyl acetate, ketones, esters and terpenes also impart the characteristic odour and flavour to this oil. Cinnamon bark oil is used in flavouring bakery foods, sauces, pickles, confectionery, soft drinks, dental and pharmaceutical preparations and also in perfumery.

Cinnamon leaf oil

Cinnamon leaf oil is produced by steam distillation of leaves yielding 0.5 to 0.7 per cent oil. It is yellow to brownish yellow in colour and possesses a warm, spicy but rather harsh odour. The major constituent is eugenol (70 to 90 per cent) while the cinnamaldehyde content is less than five per cent. The oil is used in perfumery and flavouring, and also as a source of eugenol.

Cinnamon root bark oil

The root bark contains 1.0 to 2.8 per cent oil containing camphor as the main constituent. Cinnamaldehyde as well as traces of eugenol are found in the oil, having less commercial relevance.

Plant protection

Leaf spot and dieback disease (*Colletotrichum gloeosporioides*)

On young nursery seedlings, small brown specks appear which gradually enlarge resulting in drying of the leaf. From the leaves, the infection spreads to the stem, resulting in necrosis from the apex downwards.

On old seedlings and mature trees, light and dark brown concentric zonation occurs. Spraying 1per cent Bordeaux mixture during rainy season controls the disease.

The other diseases of cinnamon include grey blight caused by *Pestalotiopsis palmarum*, sooty mould caused by *Phragmocapnius* sp. and algal leaf spot by *Cephaleuros* sp.

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CLOVE (*Syzygium aromaticum*)

Clove requires a warm humid tropical climate with an annual rainfall from 150-250 cm. It grows well from mean sea level up to an altitude of 800-900 m. Deep loam soils with high humus content and black loams of semi-forest regions with good drainage are suited for the cultivation of the crop.

Selection of site

Select partially shaded sites having adequate protection from high winds. Avoid exposed and shady locations.

Seeds and sowing

Clove is propagated through seeds obtained from fully developed fruits known as mother of clove. Collect fully developed fruits from regular bearing mother trees. Dehusk the fruits immediately after collection by soaking in water and peeling. Prepare raised nursery beds with fertile soil rich in humus under the shade of trees. Sow the seeds flat at a depth 2-5 cm and a spacing of 12-15 cm. Water the beds regularly. Seedlings can either be retained in the nursery till they attain a height of 25-30 cm when they are ready for transplanting or potted when they are six months old and transplanted after another 12-18 months.

Planting

Select 18 months old seedlings for planting. Prepare pits of size 60 cm x 60 cm x 60 cm at a spacing of 6 m x 6 m about a month in advance of planting. Allow to weather. Fill up the pits with mixture of burnt earth, compost and topsoil. Plant the seedlings during the rainy season, May-June or August-September. Provide shade and irrigation during breaks in the monsoon and summer. Banana or glyricidia may be planted to provide shade.

Clove is generally grown as a mixed crop with coffee, coconut, arecanut etc.

Manuring

Apply cattle manure or compost at the rate of 15 kg per tree per annum during May-June. The recommended fertilizer dose is $N:P_2O_5:K_2O$ @ 20:18:50 g per plant during the first year and @ 40:36:100 g per plant during the second year. Increase gradually the dose to 300:250:750 g per plant per year for a well grown tree of 15 years or more. Apply organic manures in May-June with the commencement of southwest monsoon. Apply fertilizers in two equal split doses in May-June along with the organic manures and in September-October in shallow trenches dug around the plant about 1 to 1.25 m away from the base.

Aftercultivation

Conduct weeding and intercultivation whenever necessary. Cut and remove dead and diseased branches of fully grown trees to prevent over crowding. Spray 1 per cent Bordeaux mixture to control dieback.

Harvesting and curing

The trees begin to yield from 7-8 years after planting. The stage of harvest of flower buds determine the quality of the final dried product. Buds are harvested when the base of calyx has turned from green to pink in colour. If allowed to develop beyond this stage, the buds open, petals drop and an inferior quality spice is obtained on drying.

Prior to drying, buds are removed from the stem by holding the cluster in one hand and pressing it against the palm of the other with a slight twisting movement. The clove buds and stems are piled separately for drying. Buds may be sorted to remove over ripe cloves and fallen flowers. Drying should be done immediately after the buds are separated from the clusters. If left too long in heaps, they ferment and the dried spice has a whitish shriveled appearance (khoker clove).

The traditional method of drying is by exposing them to sun in mats. The green buds are spread out in a thin layer on the drying floor and are raked from time to time to ensure the development of a uniform colour and to prevent mould formation. In sunny weather, drying is completed in 4-5 days giving a bright coloured dried spice of attractive appearance. During drying, clove loses about two-third of its original fresh green weight. When properly dried, it will turn bright brown and does not bend when pressed. The dried cloves are sorted to remove mother of cloves and khoker cloves, bagged and stored in a dry place. The stem after separation of buds is dried in a similar manner as the spice, without allowing mould formation and fermentation.

Clove bud oil

The essential oil is obtained by steam distillation of comminuted buds or whole cloves. On distillation, about 17 per cent essential oil is obtained which is a colourless or yellow liquid possessing odour and flavour characteristic of the spice. Finest oil contains 85-89 per cent eugenol. Clove bud oil is used for flavouring food and in perfumery.

Clove stem oil

Clove stem oil is obtained from dried peduncles and stem of clove buds (5-7 per cent) on steam distillation. The eugenol content of the oil ranges from 90-95 per cent. This oil possesses a coarser and woodier odour than bud oil.

Clove leaf oil

Clove leaves on distillation yield 2-3 per cent oil as a dark brown liquid with a harsh woody odour. When rectified, it turns pale yellow and smells sweeter with a eugenol content of 80 to 85 per cent.

Oleoresin

Clove oleoresin may be prepared by cold or hot extraction of crushed spices using organic solvents like acetone giving a recovery of 18-22 per cent. The oleoresin is chiefly used in perfumery and when used for flavouring it is dispersed on salt, flour etc.

Plant protection**Pests**

Infestation of shoot borer *Sinoxylon* sp. on tender shoots of young plants can be prevented by prophylactic application of carbaryl 0.15 per cent. Prune off the laterals of old trees showing dieback symptoms. Do not allow dried glyricidia and other twigs to remain in the plantation, to ensure that the beetles will not multiply on these materials and subsequently initiate infestation in cloves.

Diseases

Leaf spot, twig blight and flower bud shedding (*Colletotrichum gloeosporioides*)

Three types of symptoms are seen viz., leaf spot, twig blight and flower bud shedding. On the leaves, necrotic spots of variable sizes and shapes are noticed. Severely affected leaves wither, drop and dry up. In the nursery seedlings, dieback symptoms are seen. Extension of the symptoms from the leaves through petioles results in the infection of twigs. The affected branches stand without leaves or only with young leaves at the tips. The flower buds are attacked by spread of infection from the twigs. Shedding of flower buds occur during periods of heavy and continuous rainfall.

Management

Spraying 1per cent Bordeaux mixture at 1-1.5 month intervals reduces disease intensity, defoliation and flower bud shedding. The spraying has to be commenced just prior to flower bud formation and continued till the harvest of flower buds for effective control. Destruction of the weed *Clerodendron* from the clove garden is recommended to reduce the disease since the pathogen survives on this weed during adverse conditions.

The other diseases of clove are:

Grey blight of clove (*Pestalotia palmarum*), Leaf spot of clove (*Cylindrocladium quinqueseptatum*), Leaf spot of clove (*Alternaria citri*), Sooty mould of clove (*Phragmocapnium* sp.), Algal leaf spot of clove (*Cephaleuros* sp.) and Little leaf of clove (suspected to be due to Phytoplasma).

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VANILLA (*Vanilla planifolia*)

Vanilla is a tropical orchid requiring a warm climate with frequent rains, preferring an annual rainfall of 150-300 cm. Uncleared jungle areas are ideal for establishing vanilla plantations. In such locations, it would be necessary to retain the natural shade provided by lofty trees and to leave the soil or the rich humus layer on the top undisturbed. Vanilla is cultivated on varied type of soils from sandy loam to laterites. It requires filtered sunlight. In the absence of natural shade, trees should be grown to provide shade.

Preparation of land

Clear the land of jungle growth and prepare for planting. Being a creeper, the plant requires support up to a height of about 130-135 cm. Cuttings of *Plumaria alba*, *Erythrina lithosperma*, *Jatropha carcas* and *Glyricidia maculata* are suitable as live supports. The growth of live standard is to be adjusted to make them branch at a height of 120-150 cm to facilitate trailing of the vines and artificial hand pollination.

Time and method of planting

Vanilla is propagated by planting shoot cuttings *in situ*. Plant cuttings of 60 cm length. Longer cuttings bear earlier than shorter cuttings. Rooted cuttings as well as tissue culture derived plants can also be used for planting.

Plant the cutting with the onset of monsoon rains. Set out the cutting at a spacing of 2.7 m between plants and 1.8 m between rows in pits of size 40 cm x 40 cm x 40 cm. Trail the vines on the live supports and when they attain a height of 135 cm trail them horizontally on bamboo poles tied to vertical supports or branches of support plants in loops touching the ground.

Manuring

Being a surface rooting plant, manuring should be confined to the surface layer of soil. Provide heavy and frequent mulching to the vines. Apply 120 g of N in the form of leaf mould or FYM in two split doses in June-July and September-October.

Aftercultivation

Vanilla cannot withstand even the slightest root disturbance. Hence remove weeds from the plant base by hand weeding and use them as mulch.

Being closely planted, no intercrops are raised in a pure plantation of vanilla. But vanilla can be planted as an intercrop in coffee, coconut, arecanut etc.

Pollination, harvesting and curing

Flowering of vine commences usually by about the third year. The inflorescence is produced in the leaf axils. There is a tendency for some of the vines to maintain only vegetative growth. A light nipping off or pruning of the terminal shoots hastens flowering. Due to the peculiar structure of the flowers, self-pollination is impossible. Hence hand pollination is adopted for fruit set. Best time for pollinating the flowers is between 6 am and 1 pm and a success of 80-85 per cent can be obtained. Successful fertilization is indicated by the retention of calyx and the stigma even after four days of pollination.

The pods ripen in about 9-11 months time. Before attaining maturity the fruit is dark green in colour and when ripe yellowing commences from the tip of the pod. Collect the pods at this time, as this is the optimum time for harvesting the pod. If allowed to remain on the vine further, the pods split. Free vanillin is not present in the beans when they are harvested. They also do not have the aroma. Vanillin is developed as a result of enzyme action on a glycoside occurring during the process of curing of beans.

Harvested beans are subjected to curing which is characterized by four phases.

1. Killing or wilting beans to arrest the vegetative development in the fresh beans and initiate the enzymatic reactions responsible for the production of aroma and flavour. Killing is indicated by the development of a brown colouration of the bean.
2. Raising temperature of the killed beans (sweating) to promote the desired enzymatic reactions and to achieve rapid drying so as to prevent harmful fermentation.
3. Slow drying at ambient temperature until the beans have reached about one-third of original weight for the development of various fragrant substances.
4. Conditioning the beans by storing them in closed boxes for three months or longer to permit the full development of desired aroma and flavour.

Curing of vanilla involves immersing the beans (2-3 days after harvest) in hot water at a temperature of 63 to 65°C for three minutes for the cessation of vegetative life. After a rapid drying on woolen blankets, when the beans are still very hot, they are kept in chests lined with blankets. Next day they are spread out in sun on blanket for three to four hours and rolled up to retain the heat. Repeat this for six to eight days during which beans lose their weight, become supple and can be twisted on finger without breaking. This is followed by slow drying in the shade for a period of two to three months. Properly dried beans are kept in trunks where the fragrance is fully developed. Finally, they are graded according to size and bundled and placed in iron boxes lined with paraffin paper. The vanillin content of properly cured beans will be about 2.5 per cent.

Plant protection

The occurrence of a wilt disease caused by *Fusarium oxysporum* has been observed. For control of wilt disease adopt the following measures.

1. Remove diseased plants along with surrounding soil where the disease is observed.
2. Remove weeds around the plants.
3. Mulch the base of the vine with dry leaves before and after monsoon.
4. Avoid injury to roots during cultivation.
5. Drench soil around the base of vine with 1 per cent Bordeaux mixture.

Fungal diseases like shoot tip rot, stem and bean rot caused by *Phytophthora* sp. as well as immature bean drop are noticed. The disease affected portions are to be removed regularly and 1 per cent Bordeaux mixture should be applied on the affected plants.

Ginger is a tropical plant adapted for cultivation even in regions of subtropical climate such as the high ranges. It prefers a rich soil with high humus content. Being an exhausting crop, ginger is not cultivated continuously in the same field but shifting cultivation is practised. The crop cannot withstand waterlogging and hence soils with good drainage are preferred for its cultivation. It is shade tolerant / loving crop with shallow roots and therefore suitable for intercropping and as a component in the homesteads where low to medium shade is available.

Clear the field during February-March and burn the weeds, stubbles, roots etc. *in situ*. Prepare the land by ploughing or digging. Prepare beds of convenient length (across the slope where the land is undulating), 1 m width, 25 cm height with 40 cm spacing between the beds. Provide drainage channels, one for every 25 beds on flat lands.

Rio-De-Janeiro and Karthika (tolerant to rhizome rot and bacterial wilt).

Mark healthy and disease free plants in the field when the crop is 6-8 months old and still green. Select best rhizomes free from pest and disease from the marked plants. Handle seed rhizomes carefully to avoid damage to buds. Soak the selected rhizomes for 30 minutes in a solution of mancozeb and malathion to give terminal concentration of 0.3 per cent for the former and 0.1 per cent for the latter. Dry the treated rhizomes in shade by spreading on the floor. Store the treated rhizomes in pits dug under shade, the floor of which is lined with sand or saw dust. It is advisable to spread layers of leaves of *Glycosmis pentaphylla* (panal). Cover the pits with coconut fronds.

Examine the stored rhizomes at monthly intervals and remove the rhizomes that show signs of rotting. This will help to keep the inoculum level low. Provide one or two holes for better aeration. Treat the seed rhizomes similarly before planting also.

The best time for planting ginger is during the first fortnight of April, after receipt of pre-monsoon showers. For irrigated ginger, the best suited time for planting is middle of February (for vegetable ginger).

Plant rhizome bits of 15 g weight in small pits at a spacing of 20 cm x 20 cm to 25 cm x 25 cm and at a depth of 4-5 cm with at least one viable healthy bud facing upwards.

Seed rate 1500 kg ha⁻¹

Apply manures and fertilizers at the following rates:

FYM 30 t ha⁻¹

N:P₂O₅:K₂O 75:50:50: kg/ha/year

Full dose of P_2O_5 and 50 per cent of K_2O may be applied as basal. Half the quantity of N may be applied 60 days after planting. The remaining quantity of N and K_2O may be applied 120 days after planting.

Immediately after planting, mulch the beds thickly with green leaves @ 15 t ha⁻¹. Repeat mulching with green leaves twice @ 7.5 t ha⁻¹ first 44-60 days and second 90-120 days after planting. Grow green manure crops like daincha and sun hemp in the interspaces of beds, along with ginger and harvest the green manure crop during second mulching of ginger beds.

Aftercultivation

Remove weeds by hand weeding before each mulching. Repeat weeding according to weed growth during the fifth and sixth month after planting. Earth up the crop during the first mulching and avoid water stagnation.

Plant protection

1. For controlling shoot borer, spray dimethoate or quinalphos at 0.05 per cent
2. For controlling rhizome rot, adopt the following measures:
 - a. Select sites having proper drainage.
 - b. Select seed rhizomes from disease free areas.
 - c. Treat seed rhizomes with 0.3 per cent mancozeb.
 - d. When incidence of rhizome rot is noted in the field, dig out the affected plants and drench the beds with cheshunt compound or 1.0 per cent Bordeaux mixture or 0.3 per cent mancozeb.
 - e. Inoculation with native arbuscular mycorrhiza, *Trichoderma* and *Pseudomonas fluorescens* at the time of planting is recommended as a biocontrol measure.
3. For controlling the leaf spot disease, 1 per cent Bordeaux mixture, 0.3 per cent mancozeb or 0.2 per cent thiram may be sprayed.
4. For control of nematode in endemic area, apply neem cake @ 1.0 t ha⁻¹ at planting followed by application of neem cake @ 1.0 t ha⁻¹ at 45 days after planting (DAP).

Harvesting and processing

For vegetable ginger, the crop can be harvested from sixth month onwards. For dry ginger, harvest the crop between 245-260 days. After harvest, the fibrous roots attached to the rhizomes are trimmed off and soil is removed by washing. Rhizomes are soaked in water overnight and then cleaned. The skin is removed by scrapping with sharp bamboo splits or such other materials. Never use metallic substances since they will discolour the rhizomes. After scrapping, the rhizomes are sun dried for a week with frequent turnings. They are again rubbed well by hand to remove any outer skin.

Ginger oil

Ginger oil is prepared commercially by steam distillation of dried powdered ginger. The yield of oil varies from 1.3 to 3.0 per cent. The major use of ginger oil is as a flavouring agent for beverages, both alcoholic and non-alcoholic.

Ginger oleoresin

Oleoresin from ginger is obtained conventionally by extraction of dried powdered ginger with organic solvents like ethyl acetate, ethanol or acetone. Commercial dried ginger yields 3.5-10.0 per cent oleoresin. Ginger oleoresin is a dark brown viscous liquid responsible for the flavour and pungency of the spice.

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MANGO-GINGER (*Curcuma amada*)

It is an under exploited spice crop, which grows luxuriantly in tropical soils with good drainage. The rhizomes of mango ginger are used for preparing pickles, chutney, preserve, candy, sauce and salad and in meat and other culinary preparations. The rhizome has excellent medicinal properties and finds extensive use in the indigenous system of medicine. It is, appetizer, antipyretic, aphrodisiac and laxative. It is useful in biliousness, itching, skin diseases, bronchitis, asthma, hiccough and inflammation due to injuries. The rhizomes and roots are carminative and stomachic and in crushed pulp form they are applied over contusions, sprains and bruises for rapid healing.

Mango ginger is botanically related to neither mango nor ginger, but to turmeric (*Curcuma longa*). Morphologically mango ginger plant is similar to turmeric, but has shorter crop duration of six months. The rhizomes are pale yellow inside with lighter colour outside, have sweet smell of unripe mango when crushed. The crop comes up well in open conditions, but tolerates low levels of shade and therefore partially shaded situations can also be utilized for its cultivation. It can be well accommodated as an intercrop in coconut gardens and in rotation with other short duration crops like vegetables and also as a crop component in homesteads.

Preparation of land

Prepare the land to a good tilth during February-March subject to the availability of pre-monsoon showers. Prepare beds of convenient length, 1.2 m width, 25 cm height and 40 cm spacing between beds.

Seed material and varieties

Whole or split mother rhizomes or well developed, healthy and disease free finger rhizomes weighing 15-20

g are suitable for planting. In Kerala, local varieties are used for cultivation. Amba is a released variety from High Altitude Research Station, Pottangi, Orissa.

Season and method of planting

Plant during April with the commencement of pre-monsoon showers. Take small pits in the beds with a spacing of 25 cm x 30 cm and at a depth of 4-5 cm. Adopt a seed rate of 1500 kg ha⁻¹.

Manuring

Apply cattle manure or compost as basal dose @ 30-40 t ha⁻¹, spread over the beds and mix well. Apply N:P₂O₅:K₂O fertilizer @ 30:30:60 kg ha⁻¹. Full dose of P₂O₅ and half dose of K₂O may be applied as basal. Apply two-third dose of nitrogen 30 days after planting and remaining N and K₂O at 60 days after planting.

Mulching

Mulch the crop immediately after planting with green leaves @ 15 t ha⁻¹. Repeat mulching after 50 days with same quantity of green leaves.

Aftercultivation

The rhizomes germinate within 3-4 weeks. Remove weeds 45 days after planting and repeat if necessary. Earth up the crop after 60 days of planting.

Plant protection

Compared to the related crops ginger and turmeric, the crop is free from pests and diseases. But when large scale cultivation is taken up, the attack of shoot borer (*Conogethes punctiferalis*) causes damage to the crop. Appearance of dead heart in the field is the main symptom. To reduce the pest population, pull out the dead hearts with the larvae inside and burn it. If infestation is severe, spray dimethoate or quinalphos at 0.05 per cent.

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NUTMEG (*Myristica fragrans*)

Nutmeg requires a hot, humid climate without pronounced dry season. The soil should be rich in organic matter and well drained. The tree prefers partial shade. Sheltered valleys are the best suited. It can be grown up to about 900 m above MSL.

Variety: IISR-Viswashree

Seeds and sowing

Fully ripe tree-burst fruits are selected for raising seedlings. The fleshy rind and the mace are removed before sowing. The seeds should be sown immediately after collection. If there is any delay in sowing, the seeds should be kept in baskets filled with damp soil. The seedbeds of 100-120 cm width, 15 cm height and of convenient length may be prepared in cool and shady places. A mixture of garden soil and sand in the ratio 3:1 may be used for preparing nursery beds. Over this, sand is spread to a thickness of 2-3 cm and the seeds dibbled 2 cm below the surface at a spacing of about 12 cm on either side. Seeds germinate within 50-80 days after sowing. When the plumule produces two elongated opposite leaves, the seedlings are to be transferred from beds to polybags.

Vegetative propagation of nutmeg through epicotyl grafting is recommended for all nutmeg nurseries.

Planting

Since the nutmeg trees require shade, suitable fast growing shade trees like *Albizia*, *Erythrina* etc. are planted in advance. Banana can also be grown as a shade crop in the early stages. Pits of 90 cm x 90 cm x 90 cm are dug at a spacing of 8 m x 8 m with the onset of southwest monsoon. The pits are filled with topsoil and compost or well decomposed cattle manure and seedlings are planted.

Manuring

Apply 10 kg cattle manure or compost per seedling during the first year. Increase the quantity gradually till a well grown tree of 15 years and above receives 50 kg of organic manures per year. Apply N:P₂O₅:K₂O @ 20:18:50 g/plant during the first year. This may be doubled in the next year. Gradually increase the N:P₂O₅:K₂O dose to 500:250:1000 g/plant/year to obtain full dose from 15th year onwards.

Harvesting

Fruits are available throughout the year, but the peak period of harvest is from June to July. When fruits are fully ripe, the nuts split open. These are either plucked from the tree or allowed to drop. The two major products are nutmeg and mace. Dried nutmeg and mace are directly used as spice and also for the preparation of their derivatives.

After de-rindng the nutmeg fruit, red feathery aril (mace) is separated from pericarp. The mace is detached, flattened and sun dried on mats for 3-5 days or in artificial heat drier.

The nuts are sun dried for four to eight weeks or in artificial heat drier until kernel rattle inside the shell. They are stored in warm dry place prior to shelling.

Oleoresin

Nutmeg and mace oleoresins are prepared by extracting the ground spice with organic solvents. Yield of oleoresin is 10-12 per cent for nutmeg and 10-13 per cent for mace. Mace oleoresin possesses a fine, fresh fruity character.

Nutmeg butter

Nutmeg contains 25-40 per cent of fixed oil that can be obtained by pressing the crushed nuts between plates in the presence of steam or by extracting with solvents. The product, known as nutmeg butter, is a highly aromatic, orange coloured fat with the consistency of butter at ambient temperature.

Nutmeg oil

This is obtained as pale yellow to white volatile liquid possessing a fresh warm aromatic odour. The yield ranges from 7 to 16 per cent. The unshelled nuts are coarsely crushed in a mechanical cracker and steam distilled.

Mace oil

The mace yields 4-17 per cent colourless to pale yellow liquid possessing organoleptic properties similar to nutmeg oil. Nutmeg and mace oil are also used for flavouring.

Pests

The hard scale *Saissetia nigra* occurs on the pencil thick branches and desaps the tissues. The infested shoots invariably develop sooty mould cover. It can be controlled by spot spraying with quinalphos 0.025 per cent.

Diseases

Leaf spot and shot hole (*Colletotrichum gloeosporioides*)

Sunken spots surrounded by a yellow halo are the initial symptoms. Subsequently the central portion of the necrotic region drops off resulting in shot hole symptoms. Dieback symptoms are also observed in some of the mature branches. On young seedlings drying of the leaves and subsequent defoliation are seen. The disease can be controlled by spraying 1per cent Bordeaux mixture two or three times during rainyseason.

Fruit rot

This is caused by *Colletotrichum gloeosporioides* and *Botryodiplodia theobromae*. Water soaked lesions are seen on the fruits, the tissues of which become discoloured and disintegrated. Premature splitting of the pericarp and rotting of mace and seed are the main symptoms of the disease. The internal tissues are found rotten. The fallen fruits become enveloped with the growth of the organism. The disease can be controlled by spraying 1per cent Bordeaux mixture.

The other diseases include leaf blight (*Botryodiplodia theobromae*), leaf spot (*Alternaria citri*), sooty mould (*Phragmocapnius* sp.) and the algal leaf spot (*Cephaleuros* sp.).

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PEPPER (*Piper nigrum*)

Pepper requires a warm and humid climate. Though an annual rainfall of 250 cm is ideal for the proper growth of the crop, it can also come up well in low rainfall areas, if the pattern and distribution of rainfall are conducive. About 70 mm of rainfall within a period of 20 days may be sufficient for triggering of flushing and flowering process in the plant, but once the process is set on, there should be continuous, rainfall until fruit development starts. Any dry spell, even for a few days, within this critical period will result in substantial reduction of yield. Very long spells of dry weather are unfavourable for the crop growth.

The plant tolerates a minimum temperature of 10°C and maximum of 40°C, the optimum being 20 - 30°C. It can be grown from sea level upto an altitude of 1200 m.

Pepper prefers a light porous and well-drained soil rich in organic matter. Water stagnation in the soil, even for a very short period, is injurious for the plant. So, heavy textured soils in locations where drainage facilities are inadequate should be avoided.

Varieties

Improved varieties: Panniyur-1, Panniyur-2, Panniyur-3, Panniyur-4, Panniyur-5, Panniyur-6, Panniyur-7, Subhakara, Sreekara, Panchami, Pournami. IISR Sakthi and IISR Thevam are tolerant to Phytophthora foot rot.

Local varieties: Kottanadan, Kuthiravally, Arakulam Munda, Balankotta, Kalluvally and Karimunda.

Selection of site

Sites with slight to moderate slope are ideal for pepper cultivation, as they promote drainage. Slopes facing south are to be avoided as far as possible. When such slopes are to be used for cultivation, the young plants may be sufficiently protected from the scorching sun during summer.

Selection of mother plants

Cultivate varieties, which are proven to be highly productive. Select mother plants, which give regularly high yields and possess other desirable attributes such as vigorous growth, maximum number of spikes per unit area, long spikes, close setting of berries, disease tolerance etc. Selected mother plants should be in the age group of 5-12 years. Mark and label selected mother plants in October-November.

Raising of rooted cuttings

Pepper is propagated vegetatively from cuttings. Select runner shoots produced at the base of mother plants and keep them coiled and raised to prevent from striking roots in the soil. Separate them from the vines in February-March. The middle one-third portion of runner shoot is preferred for planting. Very tender and too hard portions of the shoots are to be avoided. The shoots are cut into pieces with 2-3 nodes in each. Two node semi-hard wood cuttings are to be planted for rooting of pepper cuttings. Leaves, if any, are to be clipped off leaving a small portion of the petioles on the stem. Satisfactory rooting and survival of cuttings (over 70 per cent) could be achieved even without any hormone treatment. Plant the cuttings in polythene bags filled with potting mixture. The potting mixture is prepared by mixing two parts of fertile topsoil, one part of river sand and one part of well rotten cattle manure. Substituting granite powder (a waste material from stone quarries) for sand in conventional potting mixture (2:1:1) is good for growth of pepper cuttings and is economical. Recommended for black pepper nurseries for large scale multiplication. Solarized potting mixture supplemented with nutrient solution (urea, superphosphate, MOP and magnesium sulphate in 4:3:2:1 ratio) and fortified with biocontrol consortia promotes growth and helps in production of disease free rooted cuttings. When polythene bags are used, sufficient number of holes (16-20) may be provided at the base to ensure good drainage. The cuttings should be planted at least one node deep in the soil. The cutting after planting should be kept under good shade. In large nurseries, pandals are to be constructed for this purpose. The cuttings are to be well protected from direct sunlight and frequent watering is recommended in the nursery to maintain a humid and cool atmosphere around the cuttings. Watering 2-3 times a day is sufficient. Heavy watering, which makes the soil slushy and causes water logging is to be avoided.

Serpentine method of propagation

Three node cuttings planted in polythene bags are kept in a corner of the nursery. When the plant develops two leaves they are trailed horizontally in polythene bags containing potting mixture kept below each tender node. Each node will be pressed into the mixture with polythene bags with 'V' shaped midribs of coconut leaves. As new shoots arise these will be trailed horizontally in polythene bags containing potting mixture. Upward growth of cutting is not arrested. Once twenty nodes get rooted first 10 bags in the rooted nodes will be separated by cutting at the inter nodes. The inter nodal stub will be pushed back into the potting mixture. These stubs also produce a second root system. Daily irrigation is to be given using a rose can. After three months it will be ready for planting in the main field. On an average 60 cuttings will be obtained in a year by this method from each mother cutting. Recommended in black pepper nurseries for large scale multiplication.

Field planting

Planting of standards is to be taken up in April-May with the onset of pre-monsoon showers. Murukku (*Erythrina indica*) Karayam or Killingil (*Garuga pinnata*), *Ailanthus* sp., subabul (*Leucaenea leucocephala*) etc. are suitable standards for growing pepper. Because of prevalence of *Erythrina* gall wasp avoid using *Erythrina* as standard. Only species tolerant to *Erythrina* gall wasp is *Erythrina variegata* (heavily thorny). In high altitude areas, dadap (*E. lithosperma*) and Silver oak (*Grevillea robusta*) can be successfully used as

standard for pepper. Seedlings of subabul and silver oak are to be planted 2-3 years before planting pepper. The cuttings of standards are to be planted in narrow holes of 40 to 50 cm depth. The spacing recommended is 3 m x 3 m on plain lands and 2 m between plants in rows across the slope and 4 m between rows on sloppy lands. The soil should be pressed well around the standards to avoid air pockets and keep the standards firm in the soil.

For planting pepper, prepare pits on the northern side of standards, 15 cm away from it. The pit size should be 50 cm x 50 cm x 50 cm. Fill the pits with a mixture of topsoil and compost or well rotten cattle manure @ 5 kg per pit. With the onset of southwest monsoon in June-July, plant 2 rooted cuttings in the pits at a distance of about 30 cm away from the standards. Press the soil around the cuttings and form a small mound slopping outward and away from the cuttings to prevent water stagnation around the plants. The growing portions of the cuttings are to be trailed and tied to the standards. Provide shade to the plants if the land is exposed and if there is a break in the rainfall. When pepper is trailed on arecanut, plant the cuttings 1.0 m away from palm and 1.5 m when coconut is used as the support. Trail the pepper vines on a temporary stake for 1-2 years. When they attain sufficient length to reach the tree trunk, remove the stake without causing damage to the vines and tie the pepper plants on to the tree trunk and trail them on it.

Management after planting

If the terrain of the land is sloppy or uneven, carry out contour bunding or terracing to prevent soil erosion. Carry out digging around the standards and vines at a radius of about 1 m from the base or in the entire plantation, twice during the year, the first at the onset of southwest monsoon and the second towards the end of northeast monsoon. Weeding around the plants is to be done according to necessity. However, in foot rot affected gardens, digging should be avoided and weeds removed by slashing. In the early stages, tie the vines to the standards, if found necessary.

Where pepper is grown in large areas, growing of cover crops like *Calapagonium muconoides* is recommended. When such cover crops are grown, they are to be cut back regularly from the base of the plants to prevent them from twining along with the pepper vines. Lowering of vines after one year's growth will promote lateral branch production.

Intercropping of pepper gardens with ginger, turmeric, colocasia and elephant foot yam is advantageous. Banana as an intercrop in yielding gardens reduces pepper yield. Therefore, this is not recommended beyond three to four years after planting of pepper vines. However, in the early years, banana provides shade to the young plants and protects them from drying up during summer months.

When pepper is grown in open places, shading and watering of the young seedlings may be done during summer months for the first 1 to 3 years according to necessity. The young plants may be completely covered with dry arecanut leaves, coconut leaves or twigs of trees until summer months are over. Mulching the basins of pepper vines during summer months is highly advantageous. Saw dust, arecanut husk and dry leaves are suitable mulching materials. Removal of unwanted terminal shoot growths and hanging shoots should be done as and when necessary.

Prune and train the standards in March-April every year to remove excessive overgrowth and to give them a proper shape. The effective height of the standard is to be limited to about 6 m. A second pruning of the standards may be done in July-August, if there is excessive shade in the garden.

Underplanting

After regular bearing for about 20 years, the vines of most varieties start declining in yield. The age of decline in yield varies with variety and agroclimatic and management factors. So underplanting should be attempted at about 20 years after planting or when a regular declining trend in yield appears. The old and senile vines can be removed 3-5 years after underplanting depending upon the growth of the young vines.

Manuring

Manuring for pepper vines is to be done in basins taken around the plant, 10-15 cm deep and 50-75 cm radius, depending up on the growth of the plants. Apply cattle manure / compost / green leaves @10 kg / plant / annum just at the onset of southwest monsoon and cover lightly with soil. It is desirable to apply lime at the rate of 500 g/vine in April-May, with the receipt of pre-monsoon showers, in alternate years.

Recommended nutrient dosage for pepper (3 years and above) is:

$N:P_2O_5:K_2O$ g/vine/year

50:50:150 (general recommendation)

50:50:200 (for Panniyur and similar areas)

140:55:275 (for Kozhikode and similar areas)

Note: Apply 1/3 dose for one year old plants and 1/2 dose for two year old plants.

The fertilizers may be applied in two split doses, the first in May-June with the receipt of a few soaking rains and the second in August-September. Apply fertilizers in a circle of radius 30 cm around the vine in the case of plants trailed on erythrina (*Nadan murukku*) or teak pole (dead standard) soil application of Zinc @ 6 kg ha⁻¹ as zinc sulphate or foliar spray of Zn @ 0.5 per cent during flowering and pin head stage of black pepper is recommended in Zinc deficient areas of black pepper cultivation for increasing the yield and quality. Application of Molybdenum @ 1 kg ha⁻¹ is recommended for areas deficient in soil molybdenum availability.

Irrigation

Irrigating pepper plants of Panniyur-1 variety at IW/CPE ratio of 0.25 from November / December till the end of March and withholding irrigation thereafter till monsoon break, increases pepper yield by about 50 per cent. The depth of irrigation recommended is 10 mm (100 litres of water per irrigation at an interval of about 8-10 days under Panniyur conditions). The water is to be applied in basins taken around the plants at a radius of 75 cm. The basins may be mulched with dry leaves or other suitable materials.

Bush pepper

For production of bush pepper, two to four node semi hard wood lateral branches are to be collected with a segment of orthotropic shoot intact and planted in the nursery for rooting during May-June. Well rooted plants are used for field planting. The rooted cuttings are to be planted at 3 pits or pots. Fertilizers can be applied @ 1.0, 0.5 and 2.0 g/pot of N, P₂O₅ and K₂O respectively at bimonthly interval. Alternatively, application of 15 g groundnut cake or 33 g of neem cake can also meet the N requirement of the crop. The bushy nature of the plant will have to be ensured by proper pruning of the viny growth. The potted plants are to be kept preferably under partial shade. It is necessary that re-potting is carried out after every two years.

Irrigating black pepper vines with 8 litres of water through drip per day during Oct - May enhances yield and quality in bush pepper with high BC ratio. Recommended for bush pepper grown as intercrop in coconut gardens.

Plant protection

Pests

For the control of pollu caused by the flea beetle *Longitarsus nigripennis*, spray any one of the following insecticides namely, dimethoate or quinalphos at 0.05 per cent concentration. The spraying is to be given at the time of spike emergence (June-July), at berry formation (September-October) and once again at berry maturing stage, if needed. It can also be controlled by spraying cypermethrin 0.01 per cent twice, first at the berry formation stage and the second one-month after the first spray (Sept-Oct.).

For controlling pepper leaf gall thrips, dimethoate 0.05 per cent may be used.

Three different types of scale insects are found infesting black pepper in high ranges of Idukki district. They are black pepper mussel scale (*Lepidosaphes piperis* Gr.) infesting all parts of vines, coconut scale (*Aspidiotus destructor* Sign) feeding from undersurface of leaf and soft scale *Marsipococcus marsupiale* Gr. confining to upper leaf surface. Infestation by mussel scale causes significant loss of yield as it affects all parts of plant including berries.

Two sprayings of dimethoate 0.05 per cent at fortnightly intervals after the harvest of berries effectively control black pepper mussel scale.

Soft scale (*Lecanium* sp.) is occasionally found to infest the foliage and vines at higher elevations. This scale insect can be controlled by spraying quinalphos 0.05 per cent. This treatment will be adequate to control the mealy bugs also. Root mealy bugs can be controlled by drenching the basins of vines with chlorpyrifos 0.075 per cent. Adequate precaution has to be taken to ensure that the insecticide solution reaches the root zone of the vines. Many of the vines infested by root mealy bugs are also likely to be infected with *Phytophthora* and nematodes. For controlling hard scale, spot application of dimethoate 0.1 per cent is recommended.

Top shoot borer can be controlled by spraying dimethoate (0.05 per cent) on the tender shoots and flushes. The spraying has to be repeated to protect newly emerging shoots and flushes.

For control of the burrowing nematode *Radopholus similis* and the root knot nematode *Meloidogyne incognita*, adopt the following measures:

- (a) Use nematode free rooted cuttings for raising new plantations.
- (b) Apply talc based formulation of *Bacillus macerans* @ 10g/vine in basins (10^6 cfu/g) at the time of planting of vines or just before the monsoon period in established plantations.

Diseases

Phytophthora (foot rot)

For controlling the disease, adopt the following management practices:

Phytosanitation

All infected or dead vines along the root system are to be removed and burnt. Wherever water stagnation is a problem, effective drainage of both surface and sub-soil is to be ensured. To avoid soil splash and consequent disease initiation and spread, a legume cover in the plantation should be ensured. Runner shoots are to be pruned or tied back to vines before the onset of monsoon. At the onset of monsoon, the branches of support trees may be lopped off to allow penetration of sunlight and avoid build up of humidity.

Apply 1 kg lime and 2 kg neem cake per standard per year as pre-monsoon dose. The application of neem cake should be four weeks after lime application.

Chemical control

For the control of *Phytophthora* foot rot, any of the following control measures can be adopted.

1. After the receipt of monsoon showers (May-June), all the vines are to be drenched over a radius 45-50 cm with 0.2 per cent copper oxychloride @ 5-10 litres per vine. This varies according to the age of the plant. A foliar spray with 1 per cent Bordeaux mixture is also to be given. Drenching and spraying are to be repeated just before the northeast monsoon. A third round of drenching may be given during October if the monsoon is prolonged.

2. After the receipt of a few monsoon showers (May-June), all the vines are to be drenched over a radius of 45-50 cm with 0.3 per cent potassium phosphonate @ 5-10 litres per vine. This varies according to the age of the plant. A foliar spray with 0.12 per cent potassium phosphonate is also to be given. A second drenching and spraying with 0.3 per cent potassium phosphonate is to be repeated just before the northeast monsoon. If the monsoon is prolonged, a third round of drenching may be given during October.

Biocontrol

Inoculate pepper vines with native arbuscular mycorrhizal fungi, *Trichoderma* and *Pseudomonas fluorescens* at the time of planting in the nursery and field and apply during the pre-monsoon period in the established plantations to control foot rot. In the field, apply the biocontrol agents around the base of the vine (see the chapter on biocontrol agents against plant pathogens).

Note: (1) All chemical control measures are prophylactic in nature and application of chemicals in advanced stages of disease will not be effective in combating the disease.

(2) In *Phytophthora* sick fields, use only chemical control measures.

Replanting / rejuvenation

Total replanting has to be undertaken in gardens where the mortality is 50-60 per cent or above. Where the mortality is below 50 per cent, timely plant protection measures as described above should be given to all the existing vines as prophylactic measure and gaps filled up. Gap filling or replanting should be taken up only after a period of one year. At the time of replanting, soil drenching with Bordeaux mixture or copper oxychloride should be given. While replanting, farmers should be encouraged to use recommended varieties.

Fungal pollu (Anthracnose)

For the control of fungal pollu or anthracnose caused by *Colletotrichum gloeosporioides*, spray 1 per cent Bordeaux mixture, once before flowering starts (late June and early July) and then at berry formation stage (late August). Minimize shade in the garden.

Foliar spray of carbendazim @ 1g l^{-1} or a formulation containing combination of carbendazim + mancozeb @ 1g l^{-1} during the month of June can effectively control the disease.

Wherever *Phytophthora* foot rot management is undertaken properly, separate control measures for pollu disease may not be necessary.

Note: Since Bordeaux mixture application for pepper is to be given mostly at a time when the monsoon is very active, it is to be ensured that a sticker is added to the fungicide. The cheapest and most effective

sticker is rosin washing soda mixture.

Rotting disease

For control of rotting disease of cuttings in the nursery, VAM and *Trichoderma* can be applied in the potting mixture. VAM inoculum consisting of infected root bits and soils can be applied @100 cc per kg of potting mixture and *Trichoderma* @ 1g kg⁻¹ of potting mixture. For the control of foliar infection apply potassium phosphonate @ 3 ml litre⁻¹ at fortnightly interval. In case, biocontrol agents are not incorporated in the potting mixture, 1 per cent Bordeaux mixture spray at weekly interval may be resorted to. When the cuttings start germination, ensure good aeration in the nursery. Heavy watering, which causes water stagnation is to be avoided. Instead, light and frequent watering should be resorted to. Remove shade as soon as continuous rain sets in.

Phyllody

In certain pockets, instead of normal spike with berries, leaf-like structures are produced. This is caused by Phytoplasma. Such vines, if noticed, must be uprooted and destroyed. Planting material should not be collected from such vines.

Stunted disease

The symptoms due to this disease include shortening of internode and narrowing of leaves with mottling. Such leaves also become leathery and deformed. This is caused by a virus. Since the disease is systemic and transmitted through planting materials, avoid collecting planting materials from such vines. Once it is noticed, uproot the vines to avoid further spread.

Waiting period of insecticide / fungicide

Dimethoate	20 days
Quinalphos	20 days
Mancozeb	30 days

Harvesting and processing

Black pepper

Black pepper of commerce is produced from whole, unripe but fully developed berries. The harvested berries are piled up in a heap to initiate browning. Then berries are detached from the stalk by threshing. Then they are spread on suitable drying floor. During sun-drying, berries are raked to ensure uniform drying and to avoid mould development. Drying the berries for 3-5 days reduces the moisture content to 10-12 per cent. The dried berries are cleaned, graded and packed in double lined gunny bags.

Blanching the berries in boiling water for one minute prior to sun drying accelerates browning process as well as the rate of drying. It also gives a uniform lustrous black colour to the finished product and prevents mouldiness of berries. Prolonged blanching should be avoided since it can deactivate the enzymes responsible for browning process.

White pepper

White pepper is prepared from ripe berries or by decorticated black pepper. Bright red berries, after harvest are detached from the stalk and packed in gunny bags. The bags are allowed to soak in slow running water for about one week during which bacterial rotting occurs and pericarp gets loosened. Then the berries are trampled under feet to remove any adhering pericarp, washed in water and then sun dried to reduce the moisture content to 10-12 per cent and to achieve a cream or white colour. White pepper is garbled, sorted and packed in gunny bags. Approximately 25 kg white pepper is obtained from 100 kg ripe berries.

Improved CFTRI method

Fully mature but unripe berries are harvested and boiled in water for 10-15 minutes to soften the pericarp. After cooling, the skin is rubbed off either mechanically or manually, washed and sun dried to obtain white pepper. Since no retting operation is involved, the product will be free from any unpleasant odour. However, white pepper produced by this method gives pepper powder of light brown colour due to gelatinisation of starch in contrast to pure white powder obtained by traditional method.

Decorticated black pepper

This is a form of white pepper produced by mechanical decortication of the outer skin of black pepper. This is generally done when white pepper is in short supply. The appearance of decorticated kernel is inferior to traditionally prepared white pepper, but is satisfactory when ground. Also the milling operation requires considerable skill to avoid excessive volatile oil loss.

Dehydrated green pepper

In this method, under-mature berries are harvested and subjected to heat treatment for inactivating the enzymes responsible for browning reaction. Then the berries are dehydrated under controlled conditions

wherein maximum retention of green colour is obtained. Dehydrated green pepper after reconstitution in water resembles freshly harvested green pepper. The advantage is that the season of availability can be extended and the berries could be stored for a year or more. Dry recovery comes to 20 per cent.

Canned green pepper

Green pepper after harvest is preserved in two per cent brine solution and the product is heat sterilized. This product has the additional advantage over dehydrated green pepper in that it retains the natural colour, texture and flavour.

Bottled green pepper

Green pepper is preserved without spoilage in 20 per cent brine solution containing 100 ppm SO₂ and 0.2 per cent citric acid. Addition of citric acid prevents blackening of berries.

Cured green pepper

To overcome the disadvantages of poor texture and weak flavour of dehydrated green pepper and the high unit weight and packing cost of canned and bottled green pepper, cured green pepper has been developed. Berries are thoroughly cleaned in water, steeped in saturated brine solution for 2-3 months, drained and packed in suitable flexible polyethylene pouches.

Freeze-dried green pepper

Most of the moisture from fresh tender green pepper is removed by freezing the berries at -30°C to -40°C under high vacuum. The colour, aroma and texture of freeze-dried green pepper are superior to sun dried or mechanically dehydrated green pepper. Freeze-dried green pepper has 2-4 per cent moisture and is very light.

Pepper oil

Black pepper is crushed to coarse powder and steam distilled to obtain 2.5 to 3.5 per cent colourless to pale green essential oil which becomes viscous on ageing. It is used in perfumery and in flavouring. Oil can also be distilled from white pepper but high price of white pepper and low oil yield do not favour its commercial production.

Pepper oleoresin

Extraction of black pepper with organic solvents like acetone, ethanol or dichloro-ethane provides 10-13 per cent oleoresin possessing the odour, flavour and pungent principles of the spice. The content of the pungent alkaloid piperine ranges from 4 to 6 per cent in dry pepper and 35 to 50 per cent in oleoresin. When freshly made, pepper oleoresin is a dark green, viscous, heavy liquid with a strong aroma. One kg of oleoresin when dispersed on an inert base can replace 15 to 20 kg of spice for flavouring purpose.

Table 23. Drying percentage and oil content of Panniyur varieties of pepper

Properties	Variety						
	P 1	P 2	P 3	P 4	P 5	P 6	P 7
Drying %	35.3	35.7	27.8	34.7	35.7	32.9	33.6
Piperine %	5.30	6.50	4.80	4.40	5.30	4.94	5.57
Oleoresin %	11.78	12.20	10.40	11.30	12.33	8.27	10.61
Essential oil %	3.31	3.40	3.12	3.12	3.80	1.33	1.50

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TURMERIC (*Curcuma longa*)

Turmeric is a tropical herb and can be grown on different types of soil under irrigated and rainfed conditions. Rich loamy soils having good drainage are ideal for the crop. It is a shade tolerant crop with shallow roots suitable for intercropping and also as a component crop in the homesteads where low to medium shade is available.

Preparation of land

Prepare the land to a fine tilth during February-March. On receipt of pre-monsoon showers in April, prepare beds of size 3 m x 1.2 m with a spacing of 40 cm between beds.

Seed material

Whole or split mother rhizomes are used for planting. Select well developed, healthy and disease free rhizomes. Treat the rhizomes in any of the copper oxychloride fungicides and store in cool, dry place or in earthen pits plastered with mud and cowdung.

Varieties

Cultivars: Tekurpetta, Sugantham, Kodur, Armoor, Alleppey.

Improved varieties: Suvarna, Suguna, Sudarshana, Prabha, Prathibha, Kanthi, Sobha, Sona, Varna, IISR Kedaram and IISR Alleppey Supreme (Resistant to leaf blotch).

Season and method of planting

Plant during April with the receipt of pre-monsoon showers. Take small pits in the beds in rows with a spacing of 25cm x 25cm. Plant finger rhizomes flat with buds facing upwards and cover with soil or dry powdered cattle manure. The seed rate is about 2000-2500 kg per ha.

Manuring

Apply cattle manure or compost as basal dose at 40 t ha⁻¹ at the time of land preparation or by spreading over the beds after planting. Apply N:P₂O₅:K₂O @ 30:30:60 kg ha⁻¹. Full dose of P₂O₅ and half dose of K₂O may be applied as basal; 2/3 dose of N may be applied at 30 days after planting; and 1/3 N and remaining K₂O may be applied 60 days after planting.

Mulching

Mulch the crop immediately after planting with green leaves @ 15 t ha⁻¹. Repeat mulching after 50 days with the same quantity of green leaves.

Aftercultivation

Weed the crop thrice at 60, 120 and 150 days after planting, depending upon weed intensity. Earth up the crop after 60 days.

Intercropping

Chilli, maize and colocasia can be grown as intercrops.

Harvesting and curing

Time of harvest depends upon variety and usually extends from January to March. Harvest early varieties at 7-8 months, medium varieties at 8-9 months and long duration varieties at 9-10 months after planting.

Improved method of processing

Cleaning: Harvested turmeric rhizomes are cleaned off mud and other extraneous materials adhering to them and subjected to curing within 2-3 days after harvest so as to ensure the quality of the end product.

Boiling: Fingers and mother rhizomes will have to be boiled separately. Boiling is usually done in MS pans of suitable size. Cleaned rhizomes (approximately 50 kg) are taken in a perforated trough of size 0.9 m x 0.55 m x 0.4 m made of GI or MS sheet with extended handle. The trough containing the rhizomes is then immersed in MS pan (1 m x 0.62 m x 0.48 m) containing clean water sufficient to immerse the rhizomes. The whole mass is boiled till the rhizomes become soft. The correct stage of cooking can be judged by piercing a wooden needle through the rhizome. If the rhizomes are properly cooked, the needle will pass through the rhizome without resistance. The cooked rhizomes are taken out of the pan by lifting the trough and draining the solution into the pan.

Drying: The fingers are then dried in the sun by spreading them as a thin layer on bamboo mats or drying floor. Artificial drying at a maximum temperature of 60°C gives a bright coloured product than that of sun drying especially for sliced turmeric.

Polishing

In order to smoothen the rough and hard outer surface of the boiled dried turmeric and also to improve its colour, it is subjected to polishing. There are two types of polishing: hand polishing and machine polishing.

Hand polishing: The method of hand polishing is simple, which consists of rubbing turmeric fingers on hard surface or trampling them under feet wrapped in gunny bags. The improved method is by using hand-operated barrel or drum mounted on a central axis, the sides of which are made of expanded metal mesh. When the drum filled with turmeric is rotated, polishing is effected by abrasion of the surface against the mesh as well as by mutual rubbing against each other as they roll inside the drum.

Machine polishing: This method consists of an octagonal or hexagonal wooden drum mounted on a central axis and rotated by power.

Turmeric oleoresin

This is obtained by the solvent extraction of the ground spice with organic solvents like acetone, ethylene dichloride and ethanol for 4-5 hours. It is orange red in colour. Oleoresin yield ranges from 7.9 to 10.4 per cent. One kg of oleoresin replaces 8 kg of ground spice.

Plant protection

No major incidence of pest or disease is noticed in the crop. Shoot borers can be controlled by spraying 0.05 per cent dimethoate or 0.05 per cent quinalphos.

Leaf spot and leaf blotch can be controlled by spraying 1 per cent Bordeaux mixture or 0.2 per cent mancozeb. If symptoms of early wilt or rhizome rot appear, drench the soil with cheshunt compound or 1 per cent Bordeaux mixture.

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CAMBOGE [KUDAMPULI] (*Garcinia gummi-gutta* var. *gummi-gutta*) (Ad hoc recommendation)

Garcinia, the camboge tree is a big sized glabrous and evergreen forest tree commonly seen in the Western Ghats of Kerala, Karnataka, and also in Sri Lanka. The tree is very much adapted to hill tops and plain lands alike. But, its performance is best in river banks and valleys. It grows well in dry or occasionally waterlogged or flooded soils.

The economic part of the plant is its mature fruit, which is highly acidic. The extract obtained from the mature fruit rind, (-) hydroxy citric acid, attracts foreign markets, for its use in medicines controlling obesity.

Planting materials

Grafts prepared through soft wood grafting or side grafting and healthy seedlings raised in the nursery are used for cultivation. If seedlings are planted, 50-60 per cent will be male; and female takes 10-12 years for bearing. Hence planting of grafts is advocated as they ensure maternal characters including early bearing tendency.

Propagation by seedlings

Selection of mother trees: Locate mother trees that give a steady annual yield with a mean fruit weight of 200-275 g, high acid and low tannin content. Collect seeds from freshly harvested and fully ripe fruits and wash in running water and spread in a thin layer under roof. By the 20th day, seeds will be ready for sowing. Sow seeds at the rate of two per bag in polybags during the month of August-September. Usually seeds start sprouting in the month of December but the sprouts become visible above the soil surface only by February. In order to avoid delayed germination, simple seed treatment methods can be employed.

Method 1: In this method, the processed seeds (after drying under shade) are given a mechanical treatment. Remove seed coats of such seeds using a sharp knife without injuring the ivory coloured cotyledon. Sow these ivory coloured cotyledons afresh in polybags at a depth of 3 cm. Germination starts in 20-25 days after sowing.

Method 2: After removing the seed coats, treat the seeds with gibberellic acid @ 250 ppm for 6 hours, and thereafter soak them in mancozeb @ 4 g per l for 2 hours. Sow the seeds in nursery bags and irrigate daily. Seeds germinate in 16-20 days.

Method 3: Second method followed by transfer of the seeds to a white polypropylene cover of size 20 cm x 25 cm along with 30-50 ml of filtered water. Tie the polybag along with the air inside tightly using a rubber band. Such seeds germinate in 10-12 days after sowing. In a polybag, about 500-750 seeds can be incubated at a time. Pick up the sprouted seeds and sow in the nursery bags kept under shade.

Keep the seedlings under shade. Irrigate them regularly on alternate days during summer months. After 3-4 months, place the seedlings under direct sunlight to trigger robust growth. At this age, apply FYM @ 50 g per bag. In six to seven months time, seedlings will be ready for planting.

Propagation by grafting

Two types of grafting methods are employed viz. soft wood grafting and approach grafting.

Soft wood grafting

Select scions only from specific elite trees regular in bearing, which produce high yield of large and quality fruits.

Collection of scion: Select straight growing, healthy, young shoots emerging from the primary branches with whorled leaf arrangement. Cut them to a length of 6-10 cm and store in polybags under humid condition. Remove leaves partly and shape the cut end to a wedge of 3-4 cm length by giving slanting cuts on two opposite sides.

Preparation of rootstock: Stock-plants having 3-4 mm stem thickness are ideal for grafting. Behead the selected plants at two nodes below the terminal bud and remove all the leaves at the graft union. Use scion and rootstock of same thickness for grafting.

Grafting: Insert the wedge of the scion into the cleft made on the rootstock and secure the graft joint firmly with a black polythene tape, 1.5-2 cm wide and 30 cm long.

Care in the nursery: Immediately after grafting, cover the plants with a transparent polypropylene cover and keep under shade. By the 30th day, grafts will establish and new leaves will start emerging. Remove the polythene cover and keep under shade. Water the grafts daily using rose-can or micro-sprinkler. Care should be taken to remove sprouts emerging from rootstock at frequent intervals. Three months after grafting the plants will be ready for planting in the main field. Just before planting in the main field, leave the grafts under open condition in 10-15 days for hardening.

Approach grafting

Here also stock plants having 3-4 mm thickness are preferred and they are brought to the place where the mother tree is located. Grafting is done as in other crops and is kept intact for 45 days by which time union occurs. Graft is detached from the mother tree in three steps. The main disadvantage is that only a limited number of grafts can be produced in this method. Forty five days after grafting, they will be ready for transferring to the main nursery for hardening. Grafts are to be watered daily using a rose-can or micro-sprinkler. Care should be taken to remove sprouts emerging from rootstock at frequent intervals. Leaf folding pests common in the nursery can be controlled by spraying with quinalphos @ 2 ml l⁻¹ at monthly intervals. One year old grafts can be used for field planting.

Planting

Prepare pits of size 1 m x 1 m at spacing of 10 m. Refill the pits with a mixture of topsoil and compost / FYM. Proper care should be given to avoid water stagnation in pits.

The plants can be raised as a pure crop or as a mixed crop in coconut and arecanut gardens. Take pits of size 0.75 m x 0.75 m x 0.75 m in hard and laterite soils; 0.50 m x 0.50 m x 0.50 m in sandy and alluvial soils, at a spacing of 4 m x 4 m for grafts and 7 m x 7 m for seedlings. In slopes of 15 per cent or more, for planting grafts, rows are spaced at 5 to 5.5 m and 3.5 m between trees in a row. For planting seedlings, rows are spaced at 8 to 12 m and at 6 to 8 m for trees in a row. Planting is generally done at the onset of monsoon showers. Under existing coconut plantation of 25 years and above, spacing shall be so adjusted that it should alternate with the palms in the rows. Under Kuttanad conditions, where bunds and channels alternate, planting can be done in between two palms. Fill the pits with topsoil and 5 kg of compost or well-decomposed cattle manure and 10 g of carbaryl 10 per cent dust, to avoid white ant attack, before planting. The graft union shall remain just above the ground level. Provide support to the young plants. One month after planting, gently remove the polythene tape around the graft union.

Management of plantation

Clean the field free of bushes and thick shades. Weed once in three months and mulch the basin with black polythene or dry leaves to avoid drying.

Manuring

Apply 10 kg cattle manure or compost per seedling / graft during the first year. Gradually increase the quantity so that a well-grown tree of 15 years and above receives 50 kg of organic manure per year. Apply N:P₂O₅:K₂O mixture at the rate of 20:18:50 g/plant during the first year. Double the dose in the second year and gradually increase it to 500:250:1000 g / plant / year at the 15th year.

Pruning

Grafts will grow fast from the second year onwards. Give strong support with casuarina poles at this stage. By fifth year, the tree will have 3 to 4 m height. At this stage, height of the plant may be maintained at 3.5 to 4 m and by seventh year at 4 to 4.5 m by pruning.

Pests and diseases

Hard scales and beetles are found to infest the crop. Hard scales desap the leaves and tender shoots. Both the adult beetles and their grubs defoliate the crop inflicting heavy loss of yield. Control these pests by spraying dimethoate @ 1 ml/l. Leaf folders are very common in the nursery against which quinalphos @ 2 ml/l may be sprayed. Incidence of hoppers is observed on grafts and large trees. This causes withering of leaves, drying up of branches and yield loss. Control them by spraying carbaryl 50 WP 2g + dichlorovos 1 ml per litre of water. Sooty mould is seen associated with hard scales. Seedling blight in the nursery stage is very common. Control it by drenching nursery bed with 1per cent Bordeaux mixture or using mancozeb @ 5 g l⁻¹. In grafts and large trees, sometimes, fungal thread blights have been observed to cause leaf and twig blight. Adopt proper pruning and spray 1per cent Bordeaux mixture or mancozeb 0.3 per cent.

Harvesting

Seedlings start bearing generally at the age of 10-12 years. Grafts start bearing from the third year onwards and will attain full bearing at the age of 12 to 15 years. Flowering occurs in January-March and fruits mature in July. There are reports of off-season bearers, which bear two times a year, i.e., during January-July and September-February. Mature fruits, which are orange yellow in colour, drop off from the tree. Harvest mature fruits manually before they fall. Immediately after harvest, wash the fruits in running water and separate the fruit rind for processing.

Processing

Separated fruit rind is first sun dried and then either smoke-dried or oven-dried at 70-80°C. In order to increase the storage life and to impart softness, mix the dried rind with common salt @ 150 g and coconut oil @ 50 ml per kg of dried rind.

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TAMARIND (*Tamarindus indica*)

(Ad hoc recommendation)

The tree is particularly well adapted to semi-arid tropical regions, but can be grown in heavy rainfall areas too, provided the soil is well drained. It is adaptable to poor soil also.

Propagation technique

It is propagated by means of seeds, grafts and budding. Healthy seeds are sown in polybags and seedlings are transplanted at 40-60 cm height. Due to erratic bearing of seedling progeny, grafts are successfully used as propagules. Side grafting, inarching and patch budding are commonly practised. Budding is done on nine month old saplings for higher success.

Cultural operations

Plants of 40-60 cm height are planted during June to November at 10 m x 10 m spacing in pits of 1 m³ size incorporated with 15 kg of FYM. Regular watering till the plants establish in the field is a must. Leader shoot is cut at 3 m above ground level to induce scaffold branches. Organic manures are generally used. Intercropping with vegetables, groundnut and sesame can be done till the fifth year.

Pests and diseases

Insects like *Tribolium castaneum* and fungi are serious problems in storage and field respectively. Spray application of Quinalphos 0.05 per cent at the time of fruiting, when infestation starts, can control the storage beetle.

Harvest and yield

Seedlings start to yield 8-10 years after planting, whereas grafts and budded seedlings give yield after 4-5 years. Stabilized yield of 250 kg/tree is obtained from 9-10 years onwards. Harvesting is done from January to April. There is also a tendency of alternate bearing as in the case of mango.

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